

MANISH KUMAR

📍 New Delhi, India

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Phone restricted during office hours due to workplace security policies please call me before 9:00 or after 17:30.

PROFESSIONAL SUMMARY

AI Engineer with over of 6 year's experience building **production-grade computer vision and secure ML systems** for NHAI & defense (DRDO) and infrastructure (NHAI/MoRTH). Skilled in **adversarial robustness** (IBM ART), **model explainability**, **Data Quality**, **trustworthiness** and **MLflow** framework, and real-time edge deployment on NVIDIA CUDA/Jetson GPUs. Currently focused on **Generative AI (LLMs, RAG, prompt engineering)** with hands-on project experience.

SKILLS

Generative AI & LLMs:

LLMs, RAG, Prompt Engineering, Conversational AI, NLP

ML/DL & Computer Vision:

PyTorch, TensorFlow, CNNs, Deep Learning, OpenCV, YOLOv4

Secure & Explainable AI:

IBM ART (Model Evasion, Extraction, Poisoning PGD, CW, FAB, Adversarial Robustness, Captum

Deployment & Backend:

NVIDIA GPUs CUDA/Jetson (Xavier NX, Nano, Orin NX), Docker, Linux, Git, REST APIs, FastAPI, Django

Programming & Data:

Python, NumPy, Pandas

PROFESSIONAL EXPERIENCE

Software - AI Engineer

GVR Technolabs Pvt. Ltd. (DRDO – Ministry of Defence Projects)

📍 New Delhi | 📅 Oct 2024 – Present

- Conducted AI/ML security assessments using **IBM ART, Dioptra, Captum, GANs, Docker, and Linux**, focusing on adversarial robustness, explainability, and model trustworthiness.
- Performed **evasion and poisoning attacks** against **ResNet, MobileNet, VGG19, and DenseNet** models trained on the **ImageNet** dataset to identify security vulnerabilities and evaluate model resilience.
- Evaluated the **security, safety, and performance** of **Generative AI and LLM** systems, including **Hugging Face, Gemma**, and locally deployed models, while automating testing and validation workflows.

R&D Engineer – AI/ML

VaaaN Infra Pvt. Ltd. (NHAI/MoRTH Smart Highway Projects)

📍 Haryana/Faridabad | 📅 Aug 2022 – Oct 2024

- Developed and optimized **YOLO-based vehicle detection and classification models** for smart highway applications.
 - Built custom datasets from highway camera feeds and trained models using **YOLO, OpenCV, and Darknet**.
 - Implemented real-time traffic analytics for **vehicle detection, classification, and incident monitoring**.
 - Deployed AI models **on NVIDIA CUDA Jetson (Nano, Xavier NX, Orin NX)** and GPU-based server environments.
 - Developed monitoring and analytics dashboards **using React.js, Django, and REST APIs**.
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Earlier Experience

IT Administrator & System Engineer (2018 – 2022)

- Worked on Linux systems, networking, and infrastructure support
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PROJECTS

AI ChatBot (LLM-based)

- Built conversational AI system using **LLMs** for context-aware responses and basic task automation.

Adversarial Robustness Framework

- Testing Robustness of CNN model with ImageNet dataset using the framework of **IBM ART** with different attack methods (**PGD, APGD, Evasion, Poisoning, AutoAttack**), and **MLflow** to display the accuracy of models etc..

VIDS & ATCC Systems

- Developed in **Django real-time vehicle detection, classification, and incident monitoring systems** using **YOLO, OpenCV, and Darknet**.
 - Trained custom **object detection models** on highway **camera datasets** and deployed them on **NVIDIA CUDA GPU** edge devices for smart highway applications.
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EDUCATION

Master of Computer Applications (MCA)

Swami Vivekanand Subharti University — 2022

B.Sc. Information Technology

Maharishi University — 2018

CERTIFICATIONS

- Neural Networks and Deep Learning – DeepLearning.AI
- Python for Data Science & AI — IBM
- AWS Certified DevOps Engineer – Professional
- Linux Administration Bootcamp